MIL-P-60290C(MU)
18 June 1970
SUPERSEDING
MIL-P-60290B(MU)
2 September 1969

MILITARY SPECIFICATION

PHOTOMULTIPLIER POWER SUPPLY: 10540714

1. SCOPE

1.1 This specification covers a solid state power supply having an adjustable negative high voltage d.c. output. This power supply is a modular component of the Range Finder, Laser, AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-I-46058	Insulating Compound, Electrical, for Coating Circuit Assemblies

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10540501	Range Finder, Laser, AN/GVS-3
F10540705	Housing
C10540714	Photomultiplier Power Supply
C10540746	Connector
C10540747	Connector

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PACKAGING DATA SHEET

10540714 Packaging of Photomultiplier Power Supply

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The power supply shall be manufactured in accordance with Drawing C10540714 and drawings pertaining thereto.

3.3 General specification.- The power supply shall meet the following requirements, where applicable, of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Electrical and electronic assemblies
d. Soldering electrical connections
e. Part identification and marking
f. Workmanship

3.3.1 Electronic workmanship.- In addition to meeting the workmanship requirements of MIL-F-13926, the power supply shall meet the electronic workmanship requirements of 3.3.1.1 through 3.3.1.11.

3.3.1.1 Etched circuit board.- The conductor pattern on the etched board shall be free from blisters, wrinkles, cracks, holes, dents, scratches, and all other defects that will affect serviceability. The pattern shall be free of loosely adhering feather edges of metal. In no case shall a finished line width be more than 0.010 inch narrower than that specified on the printed circuit master. In no case shall a conductor line width be less than 0.020 inch.

3.3.1.2 Spacing of parts.- Unless otherwise specified on the applicable drawing, all parts shall be mounted with at least one point of the component in contact with the board. The component shall not be raised more than 0.031 inch from the board at any point on its contacting surface.

3.3.1.3 Symbols and markings.- Unless physically impossible, all parts shall be mounted on the board in such a manner that the proper identification markings of each part is visible.

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3.3.1.4 Transistors.- Transistors are to be mounted perpendicular to the circuit board, with spacers between transistors and the circuit board with no linear or angular displacement resultant from engagement and securing of the screws between the heat sinks and the power supply cover. Mounted transistors utilizing heat sink caps shall be of even height. The height shall be such that the top of all transistor heat sink caps shall contact the heat sink (supply case cover) so as to neither produce undue downward pressure on the transistors nor tend to lift the heat sink caps from the transistor when the screws secure the transistor heat sink caps to the supply case cover heat sink.

3.3.1.5 Bend radius.- The minimum bend radii for formed leads shall be one diameter. The maximum shall be two diameters or 0.062 inch, whichever is smaller.

3.3.1.6 Component leads.- All leads shall be covered with sleeving so that no more than 1/32 inch bare lead is exposed. Leads of transistors mounted with a spacer directly into a printed circuit board shall have 1/16 inch of lead from the transistor base to any soldered point. Transistors mounted with a spacer require no sleeving on the leads. Component leads shall be preformed to suit mounting holes. There shall be visible evidence of leads passing completely through the circuit board. The leads shall not protrude more than 3/64 inch including solder beyond the side of the circuit board opposite the component.

3.3.1.7 Heat sensitive component.- The use of heat shunts or cooling devices shall be mandatory when hand soldering components that are subject to damage by heat.

3.3.1.8 Cleaning.- After soldering operations have been completed, the assembly shall be thoroughly cleaned and rinsed. The cleaning fluid shall have a gentle flow, at no time exceeding a pressure of 5 pounds per square inch (psi). The temperature of the cleaning solution shall not exceed 125°F. After cleaning, there shall be no residual flux, surface contamination, or detrimental staining on the assembly. Cleaned assemblies shall not be allowed to become contaminated after cleaning, nor shall markings become illegible.

3.3.1.9 Drying.- The assembly may be dried by exposure to room temperature without the aid of auxiliary heating or air circulating equipment. If other than the room temperature drying method is used, the assembly shall be dried in a circulating air oven, the temperature of which shall not exceed 165°F. When compressed air or nitrogen is used for drying parts, the pressure applied to the parts shall not exceed 5 psi. The air or nitrogen shall be filtered to meet the following requirements:

a. Permissible dust particle size not to exceed 6 microns
b. Permissible moisture content limited to 2.12 grains per cubic foot (dew point 32°F maximum)
c. Dust count shall not exceed 1000 particles per cubic foot

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3.3.1.10 Coating.

3.3.1.10.1 Masking.- Prior to coating with insulating compound MIL-I-46058, the high voltage output connector 17J1 and the adjustment screw of the potentiometer 17R5 and the sides and top of the transistors shall be masked to protect them from being coated. Subsequent to coating and when masking is removed, the output terminals, the 17R5 potentiometer adjustment screw and the transistor sides and top shall be clean and free of any coating. The transistor heat sinks, when installed, shall make metal-to-metal contact on all intended surfaces.

3.3.1.10.2 Coating, insulating compound.- The printed circuit board assembly, with electrical components installed, shall be coated with an insulating compound. The insulating compound shall be Insulating Compound, Electrical, Type PUR, Polyurethane Resin conforming to MIL-I-46058. The exterior wire harness shall not be coated with the resin.

3.3.1.10.3 Cured compound.- The cured compound shall represent a smooth continuous surface, and shall pass from the mounted components to the circuit board with no break at their junction. Except for areas masked, all surfaces shall be covered with a coating of 0.001 to 0.006 inch thickness. The cured compound shall meet the following requirements:

a. Compound shall be free of visible dust particles or other foreign matter.
b. It shall be smooth and uniform and have no globules or bare spots.
c. It shall be substantially free of bubbles.
d. There shall be no blisters or cracks.
e. It shall be transparent and shall not affect marking legibility.
f. It shall be capable of being removed in local areas for replacement of components by means of a soldering iron or similar method.
g. Applied to a test board of 1/16 inch thickness, the compound shall not rupture or separate from the board, when the board is bent into an arc of twelve inches radius.

3.3.1.11 Wire termination.- When soldering wires to terminals or solder points a small gap of 1/32 inch minimum to 1/16 inch maximum shall be left between the wire insulation and the solder build up. Burned, frayed, or otherwise damaged wire insulation shall not be acceptable. Stripping of insulation shall be accomplished without disarranging cutting or otherwise damaging any strand of the conductor. Heat shall not be used to remove the outer jacket from shielded wire. Only mechanical methods of removal shall be permitted.

Exception: The hot wire stripping method may be used where the wire or cable primary insulation and jacket are made of Teflon (TFE) or equal.

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3.4 Vibration.- The power supply shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second (cps). The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequencies. Subsequent to vibration, the power supply shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.10.

3.5 Shock.- The power supply shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half sine wave with a time duration of 11 ± 1 milliseconds. The peak amplitude of each shock impulse shall be 75g's. Subsequent to shock, the power supply shall show no evidence of physical damage, and shall meet the requirements of 3.7 through 3.10.

3.6 Storage temperatures.- The power supply shall withstand thermal stabilization at temperatures of -80°Fahrenheit (F) and +160°F. Subsequent to exposure to the temperatures specified herein, the power supply shall show no evidence of physical damage, and shall meet the requirements of 3.7 through 3.10.

3.7 Performance.- The power supply shall meet the requirements of 3.7.1 to 3.7.5 inclusive for a period of time no less than 3 minutes during which the input voltage is constantly applied and the power supply is turned on and turned off for alternate 5 second intervals. The input voltage and the turn on and turn off signals shall be as specified below: (see 3.7.1, 3.7.2 and 3.7.3)

(a) Input voltage - Pin A Positive and Pin B Negative (Ground)
(b) Turn on signal: A single 3.5 ± 0.5 volt positive pulse with respect to ground having a time duration of at least 100 microseconds applied to Pin E with Pin D ground.
(c) Turn off signal: A single 1.25 ± .25 volt negative going pulse with respect to ground having a time duration of at least 150 microseconds applied to Pin C with Pin D ground.

When the power supply is turned on, the output voltage shall be developed across an 8 megohm load connected across the output connector 17J1 with the center conductor of 17J1 being the negative side of the output voltage.

3.7.1 Normal temperature operation - output voltage (Eo) adjustment.- The power supply shall be adjustable using 17R5 to produce negative output voltages from less than 1400 Vdc to greater than 1800 Vdc when the input voltage (Ei) is 26.5 ± 0.1 Vdc and the power supply is thermally stabilized at a temperature between +70° and +80°F.

NOTE: At this time potentiometer 17R5 shall be set for an Eo of the 1800 Vdc nominal and shall not be changed or sealed. All other functional requirements shall be met at this setting.

3.7.2 Low temperature operation (Eo vs Ei).- The Eo shall be not less than 1775 Vdc with an Ei of 21 ± 0.1 Vdc, when the power supply is thermally stabilized at -65°F.

3.7.3 High temperature operation (Eo vs Ei).- The Eo shall not exceed 1825 Vdc, with an Ei of 30 ± 0.1 Vdc, when the power supply is thermally stabilized at a temperature of +155°F.

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3.7.4 Input current (Ii).- At no time shall the Ii drawn by the power supply exceed 180 milliamperes.

3.7.5 Voltage spikes.- At no time shall voltage spikes in the Eo exceed 5.0 volt, peak-to-peak.

3.8 Electrical continuity.- The electrical wiring of the power supply shall conform to the electrical schematic, Drawing D10540825.

3.9 Final voltage adjustment setting.- Subsequent to meeting the requirements of 3.7 through 3.8 inclusive, the 17R5 potentiometer shall be positioned to produce a nominal output voltage of -1600 Vdc when the input voltage is 26.5 ± 0.1 Vdc and the power supply is thermally stabilized at a temperature between +70° and +80°F.

3.10 Interface requirements.- The power supply shall interface with:

(a) Mounting pad and holes located on Housing, 10540705.
(b) Connector, 10540747 (13J3).
(c) Connector, amphenol, type 5116-037550 (12P1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure

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that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% DEFECTIVE		
101. Electronic workmanship	3.3.1	Visual
102. Etched circuit board	3.3.1.1	SME; Visual
103. Spacing of parts	3.3.1.2	SME; Visual
104. Symbols and markings	3.3.1.3	Visual
105. Transistors	3.3.1.4	Visual
106. Bend radius	3.3.1.5	SME; Visual
107. Component leads	3.3.1.6	SME; Visual
108. Heat sensitive components	3.3.1.7	Visual
109. Masking	3.3.1.10.1	Visual
110. Cured compound	3.3.1.10.3	SME; Visual
111. Wire termination	3.3.1.11	SME; Visual

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MINOR: NONE

NOTE: The above characteristics shall be inspected at in-process intermediate inspection stations.

TABLE II - CLASSIFICATION OF DEFECTS
(100% INSPECTION)

		REQUIREMENT	TEST PROCEDURE
112.	Output voltage (Eo) adjustment	3.7(b), 3.7(c), 3.7.1	4.6.2.1
113.	Input current (Ii)	3.7.4	4.6.2.2
114.	Voltage spikes	3.7.5	4.6.2.3
115.	Electrical continuity	3.8	4.6.3
116.	Final voltage adjustment setting	3.9	4.6.4
117.	Interface requirements	3.10	4.6.5

MINOR: NONE

NOTE: The tests in Table II shall be conducted at a temperature between +70°F and +80°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Cleaning	3.3.1.8
Drying	3.3.1.9
Coating, insulating compound	3.3.1.10.2

4.4 Special sampling.

4.4.1 General.- Three power supplies shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

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TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable dwgs & specs - visual
302.	Fabrication	3.2	Applicable dwgs & specs - visual
303.	General specification	3.3	MIL-F-13926 - visual

NOTE: The tests in Table III shall be conducted at a temperature between +70°F and +80°F.

4.4.2 Environmental.- Three power supplies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV and shall then meet the requirements and tests in Table II.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperature (-80°F & +160°F)	3.6	4.6.1.1
305.	Low temperature operation (-65°F)	3.7.2	4.6.1.2
306.	High temperature operation (+155°F)	3.7.3	4.6.1.3
307.	Shock (between +70°F and +80°F)	3.5	4.6.1.4
308.	Vibration (between +70°F and +80°F)	3.4.	4.6.1.5

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

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4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2 the following standard test equipment shall be utilized in the performance of the applicable tests specified in 4.6:

	NOMENCLATURE	DESCRIPTION
1.	Signal source	Capable of providing a single positive pulse of the amplitude and duration specified in 3.7(b).
2.	Signal source	Capable of providing a single negative pulse of the amplitude and duration specified in 3.7(c).
3.	D.C. power source	Range of 0-30 volts, current rating 200MA.
4.	Milliammeter	Range of 0-200 milliamperes with 2% full scale accuracy.
5.	Voltmeter D.C.	Range 0-3000 volts with an accuracy of at least 1% and sensitivity of at least 20,000 ohms per volt.
6.	Oscilloscope	Rise time 100 nanoseconds or less, input impedance of 100 megohms or greater including probe impedance, time base of 10 microseconds per centimeter, capable of measuring ripple voltage at a 2 Kv D.C. level.

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4.6 Test methods and procedures.

4.6.1 Environmental.- The power supply shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the power supply shall be subjected to the tests specified in 4.6.2 through 4.6.5. Potentiometer 17R5 shall be adjusted as specified in 4.6.2.1 prior to initiation of the following environmental tests:

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.6.

4.6.1.2 Low temperature operation (Eo vs Ei).- The power supply shall be thermally stabilized at the low temperature specified in 3.7.2 utilizing the procedures specified in 4.6.1.1. All test equipment shall be connected to the test points as specified in 4.6.2 and 4.6.2.1. Adjust the power source Item 3 in 4.5.3, to the voltage specified in 3.7.2. Apply the turn-on and turn-off signals. The measured output voltage shall be as specified in 3.7.2.

4.6.1.3 High temperature operation (Eo vs Ei).- The power supply shall be thermally stabilized at the high temperature specified in 3.7.3 utilizing the procedures specified in 4.6.1.1. All test equipment shall be connected to the test points as specified in 4.6.2 and 4.6.2.1. Adjust the power source Item 3 in 4.5.3 to the voltage specified in 3.7.3. Apply the turn-on and turn-off signals. The measured output voltage shall be as specified in 3.7.3.

4.6.1.4 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the wave form, time duration and peak "G" value specified in 3.5.

4.6.1.5 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The power supply shall be firmly mounted to the vibration table. The vibration equipment shall provide the conditions of acceleration amplitude, frequency and duration specified in 3.4. The vibration test shall then be performed in accordance with Method 514 Procedure 1, Part 1, of MIL-STD-810.

4.6.2 Performance.- Connect the positive signal source Item 1 in 4.5.3 to the power supply Connector Pins "E" and ground. Connect an 8 megohm ± 5% 1 watt resistor between center contact of the output and ground. Connect the negative signal source Item 2 in 4.5.3 to pins "C" and ground. Energize the power supply as specified in 3.7. The turn-on signal shall be measured between pins "E" and "D" with the oscilloscope Item 6 and shall meet the requirement of 3.7(b). The turn-off signal shall be measured between pins "C" and "D" and shall meet the requirement of 3.7(c).

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4.6.2.1 Normal temperature operation - Output voltage (Eo adjustment).- Connect the dc power source Item 3 in 4.5.3 to the power supply Connector Pins "A" and "B" with the positive terminal to Pin "A". Connect the dc voltmeter, Item 5 in 4.5.3, across the 8 megohm resistor at the output terminal J with the negative lead connected to the center conductor. Apply the turn-on and turn-off signals specified in 3.7(b) and 3.7(c) at the time intervals specified in 3.7. Measure and adjust the potentiometer 17R5 and power source, Item 3 in 4.5.3, as specified in 3.7.1. The output voltage shall meet the requirements of 3.7.1 when thermally stabilized at the temperatures specified. Adjust potentiometer 17R5 to the nominal output volaage (Eo) specified in 3.7.1 (Note). This setting shall remain unchanged for all subsequent tests.

CAUTION: All adjustments of 17R5 must be made with no voltage applied between Pins "A" and "B".

4.6.2.2 Input current (Ii).- The input current shall be measured with a milliammeter Item 4 in 4.5.3 connected to the power supply in series with the power source for all tests. The value of the current shall not exceed the limits specified in 3.7.4.

4.6.2.3 Voltage spikes.- The voltage spikes shall be measured with the oscilloscope Item 6 in 4.5.3 connected across the 8 megohm resistor at the output terminals for all the tests. The voltage spikes shall not exceed the limits specified in 3.7.5.

4.6.3 Electrical continuity.- Continuity shall be inspected by standard measuring equipment (SME) and visually examined for compliance with 3.8.

4.6.4 Final voltage adjustment setting.- Thermally stabilize power supply as specified in 3.9. All test equipment shall be connected to the test points as specified in 4.6.2 and 4.6.2.1. Adjust the power source Item 3 in 4.5.3 to the voltage specified in 3.9. Apply the turn-on and turn-off signals. The measured output voltage shall be as specified in 3.9. See "Caution", note in 4.6.2.1.

4.6.5 Interface requirements.- Perform this test using two test housings. One housing designed to the maximum limit of each interface with a tolerance of 10% of total limit, and one designed to the minimum limit of each interface with a tolerance of 10% of total limit. Do not extend any interface beyond maximum or minimum limits. All pertinent interfaces shall be provided as depicted on Drawing F10540705 and F10540501. Power supply shall be capable of being properly positioned and secured to both test housings. The power supply Connector C10540746 shall be engaged to a test connector with pertinent interfaces conforming to requirements of Drawing C10540747. All pins of power supply connector shall engage test connector receptacle to a depth of at least .09 and continuity with all test connector pins shall be indicated. The power supply Receptacle C10540805 shall be connected to test connector conforming to amphenol type 5116-037550. Continuity between receptacle and test connector shall be indicated when fully engaged.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10540714. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- This power supply produces a negative high voltage for the photomultiplier tube of the Range Finder, AN/GVS-3.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specified
b. Applicable packaging data sheet
c. Level of protection required
d. Provisions for First Article testing

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A248

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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Department of the Army

OFFICIAL BUSINESS

Commanding Officer
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